

GLA UNIVERSITY

CSE – 3rd Year, 5th Semester

Application Build and Release Automation Lab

ASSIGNMENT – 3

Linux and Git Version Control

Subject Code: BCSE1522

Maximum Marks: 30

Scenario

A software company named **GLA Digital Solutions** is developing an **Online Library Management System**. The development team uses Linux and Git to manage project files and collaborate on application features.

Complete the following tasks.

Task 1 – Project Workspace

Create:

`<FirstName>-GLA-LibrarySystem`

```
MINGW64/c/Users/ADITYA/OneDrive/desktop/Kushagra-Gla-LibrarySystem
ADITYA@Kush MINGW64 ~/OneDrive
$ ls
Attachments/ PARIDHI_SHUKLA.pdf
Desktop/ 'Personal Vault.lnk' Bash
Documents/ Pictures/
'Getting started with OneDrive.pdf' SIH-BLUEPRINT.pdf mkdir Kushagra-Gla-LibrarySys
IdeaThon-Final-Draft.pdf Sih_rule_book.pdf cd Kushagra-Gla-LibrarySystem
Kushagra_agarwal-12-04-26.pdf desktop.ini

ADITYA@Kush MINGW64 ~/OneDrive
$ cd desktop
Check current location:

ADITYA@Kush MINGW64 ~/OneDrive/desktop
$ mkdir Kushagra-Gla-LibrarySystem
Bash

ADITYA@Kush MINGW64 ~/OneDrive/desktop
$ cd Kushagra-Gla-LibrarySystem/
pwd

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem
$ pwd
Expected:
/c/Users/ADITYA/OneDrive/desktop/Kushagra-Gla-LibrarySystem

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem
$
```

Inside it create:

- src
- docs
- config
- backup

```
MINGW64/c/Users/ADITYA/OneDrive/desktop/Kushagra-Gla-LibrarySystem
Kushagra_agarwal-12-04-26.pdf desktop.ini
ADITYA@Kush MINGW64 ~/OneDrive
$ cd desktop
Screenshot 1: pwd

ADITYA@Kush MINGW64 ~/OneDrive/desktop
$ mkdir Kushagra-Gla-LibrarySystem

ADITYA@Kush MINGW64 ~/OneDrive/desktop
$ cd Kushagra-Gla-LibrarySystem/

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem
$ pwd
Bash
/c/Users/ADITYA/OneDrive/desktop/Kushagra-Gla-LibrarySystem

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem
$ mkdir src docs config backup
2. Required folders created

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem
Check:
$ ls
src/ docs/ config/ backup/

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem
$
```

Create:

library_info.txt

Include:

- Name
- Roll Number
- Subject
- Project Name
- Semester

```

MINGW64/c:/Users/ADITYA/OneDrive/desktop/Kushagra-Gla-LibrarySystem
ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem
$ pwd
/c:/Users/ADITYA/OneDrive/desktop/Kushagra-Gla-LibrarySystem
ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem
$ nano library_info.txt
ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem
$ mkdir src docs config backup
ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem
$ ls
backup/  config/  docs/   src/
ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem
$ nano library_info.txt
ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem
$ cat library_info.txt
Name : Kushagra
Roll Number: 2415001887
Subject : Applicaton Build and Realease Automation lab
Project Name : Online Lib management system
Semester: 5th sem
Save:
ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem
$

```

Linux commands for:

- `pwd`
- directory navigation
- listing files
- displaying file contents
- displaying directory structure

```
MINGW64:/c/Users/ADITYA/OneDrive/desktop/Kushagra-Gla-LibrarySystem
Subject : Application Build and Release Automation lab
Project Name : Online Lib management system
Semester: 5th sem

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem
$ tree
bash: tree: command not found

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem
$ sudo apt install tree
Sudo is disabled on this machine. To enable it, go to the Developer Settings page in the Settings app

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem
$ find . -maxdepth 2 -print
.
./backup
./config
./docs
./library_info.txt
./src
```

```
MINGW64:/c/Users/ADITYA/OneDrive/desktop/Kushagra-Gla-LibrarySystem

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem
$ cd src

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem/src
$ pwd
/c/Users/ADITYA/OneDrive/desktop/Kushagra-Gla-LibrarySystem/src

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem/src
$ cd ..

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem
$ pwd
/c/Users/ADITYA/OneDrive/desktop/Kushagra-Gla-LibrarySystem

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem
$ cd doc
bash: cd: doc: No such file or directory

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem
$ cd docs/

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem/docs
$ pwd
/c/Users/ADITYA/OneDrive/desktop/Kushagra-Gla-LibrarySystem/docs

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem/docs
$ cd ..

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem
```

Task 2 – Git Repository

1. Initialize Git.
2. Configure Git username/email.
3. Check status.
4. Modify `library_info.txt`.
5. Add a project objective and description.
6. Check the changes.

7. Stage the modified file.
8. Create a second commit.
9. Display commit history.

```
ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem
$ cd ..
ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem
$ pwd
/c/Users/ADITYA/OneDrive/desktop/Kushagra-Gla-LibrarySystem
ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem
$ cd doc
bash: cd: doc: No such file or directory
ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem
$ cd docs/
ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem/docs
$ pwd
/c/Users/ADITYA/OneDrive/desktop/Kushagra-Gla-LibrarySystem/docs
ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem/docs
$ cd ..
ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem
$ git init
Initialized empty Git repository in C:/Users/ADITYA/OneDrive/Desktop/Kushagra-Gla-LibrarySystem/.git/

Task 2 – Git Repository
6 Marks

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (master)
$ git config --global user.name "Kushagra-Agarwal"
ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (master)
$ git config --global user.email "kushagra2004"
ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (master)
$ git config --global user.name "Kushagra-Agarwal"
ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (master)
$ git config --global user.email "kushagra2004.agrw@gmail.com"
ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (master)
$ git status
On branch master
No commits yet
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    library_info.txt

nothing added to commit but untracked files present (use "git add" to track)
ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (master)
$ git add .
```

```
ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (master)
$ git status
On branch master
No commits yet
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    library_info.txt

nothing added to commit but untracked files present (use "git add" to track)
ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (master)
$ git add .
warning: in the working copy of 'library_info.txt', LF will be replaced by CRLF the next time Git touches it
ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (master)
$ git status
On branch master
No commits yet
Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   library_info.txt

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (master)
$ git commit -m "Initial project setup"
[master (root-commit) af75b67] Initial project setup
1 file changed, 5 insertions(+)
create mode 100644 library_info.txt

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (master)
$ nano library_info.txt
ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (master)
$ git diff
warning: in the working copy of 'library_info.txt', LF will be replaced by CRLF the next time Git touches it
diff --git a/library_info.txt b/library_info.txt
index 4d8ed34..0bbaa18 100644
--- a/library_info.txt
+++ b/library_info.txt
@@ -1,5 +1,4 @@
-name : Kushagra
-roll Number: 2415001887
-subject : Application Build and Release Automation Lab
-project Name : Online Lib management system
-semester: 5th sem
+name : Kushagra Roll Number: 2415001887 Subject : Application Build and Release Automation Lab Project Name : Online Lib management system Semester: 5th sem
+Project Objective: to develop an online library management system for managing books, users and library operations. Project Description: The system will provide functionalities for managing library books, tracking book availability and maintaining library-related information.

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (master)
$
```

```
MINGW64/c/Users/ADITYA/OneDrive/desktop/Kushagra-Gla-LibrarySystem

--- a/library_info.txt
+++ b/library_info.txt
@@ -1,5 +1,4 @@
-Name : Kushagra
-Roll Number: 2415001887
-Subject : Application Build and Release Automation lab
-Project Name : Online Lib management system
-Semester: 5th sem
+Name : Kushagra Roll Number: 2415001887 Subject : Application Build and Release Automation lab Project Name :
+Online Lib management system Semester: 5th sem Project Objective: To develop an efficient Online Library
+management System for managing books, users and library operations. Project Description:
+the system will provide functionalities for managing library books, tracking book availability and maintaining libra
ry-related information.

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (master)
$ git commit -m "Add project objective and description"
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   library_info.txt

no changes added to commit (use "git add" and/or "git commit -a")

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (master)
$ git log --oneline
af75b67 (HEAD -> master) Initial project setup

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (master)
$
```

```
MINGW64/c/Users/ADITYA/OneDrive/desktop/Kushagra-Gla-LibrarySystem

af75b67 (HEAD -> master) Initial project setup
Expected:
ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (master)
$ git branch
* master

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (master)
$ git branch -M main
book-feature
main

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (main)
$ git brach book-feature
git: 'brach' is not a git command. See 'git --help'.

The most similar command is
branch

3. Book-feature par switch karo

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (main)
$ git branch book-feature
Bash

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (main)
$ git branch
git switch book-feature
book-feature
* main

Check:
ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (main)
$ git switch book-feature
M library_info.txt
Switched to branch 'book-feature'

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (book-feature)
$ git branch
git branch
* book-feature
main

Expected:
ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (book-feature)
$
```

```
MINGW64/c:/Users/ADITYA/OneDrive/desktop/Kushagra-Gla-LibrarySystem
ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (main)
$ git branch book-feature

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (main)
$ git branch
* book-feature
* main      The Book Management Module is responsible for managing books in the Online Library Management System.

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (main)
$ git switch book-feature
Switched to branch 'book-feature'

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (book-feature)
$ git branch
* book-feature
* main      1. Add new books to the library.
            2. Update existing book information.
            3. Search books by title, author, or book ID.
            4. Delete books from the library.
            5. Track book availability.

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (book-feature)
$ nano src/book.txt
Save:
ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (book-feature)
$ cat src/book.txt
Book Management Module

The Book Management Module is responsible for managing books in the Online Library Management System.

Features:
1. Add new books to the library.
2. Update existing book information.
3. Search books by title, author, or book ID.
4. Delete books from the library.
5. Track book availability.

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (book-feature)
$ |
```

```
MINGW64/c:/Users/ADITYA/OneDrive/desktop/Kushagra-Gla-LibrarySystem
$ git branch
* book-feature
* book-feature
* main      main

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (book-feature)
$ nano src/book.txt

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (book-feature)
$ cat src/book.txt
Book Management Module

Return to main

The Book Management Module is responsible for managing books in the Online Library Management System.

Features:
1. Add new books to the library.
2. Update existing books information.
3. Search books by title, author, or book ID.
4. Delete books from the library.
5. Track book availability.

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (book-feature)
$ git add src/book.txt
warning: in the working copy of 'src/book.txt', LF will be replaced by CRLF the next time Git touches it

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (book-feature)
$ git commit -m "Add Book Management Module"
[book-feature 080e1ac] Add Book Management Module
1 file changed, 10 insertions(+)
create mode 100644 src/book.txt

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (book-feature)
$ git branch
* book-feature
* book-feature
* main      * main

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (book-feature)
$ git switch main
Deletion of directory 'src' failed. Should I try again? (y/n) y
Deletion of directory 'src' failed. Should I try again? (y/n) n
M      library_info.txt
Switched to branch 'main'

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (main)
$ |
```

TASK 4 — Merge Conflict (8 Marks)

Task 3 – Feature Branch

Develop a **Book Management Module**.

Create branch:

`book-feature`

Inside `src`, create:

`book.txt`

The file must describe the Book Management Module and contain at least three features.

Stage and commit the changes.

Demonstrate:

- branch creation
- branch switching
- current branch
- all branches
- returning to main


```
MINGW64/c:/Users/ADITYA/OneDrive/desktop/Kushagra-Gla-LibrarySystem
$ git branch
* book-feature
main
extensions Help
ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (book-feature)
$ nano src/book.txt
text Arial 11 B I U A
ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (book-feature)
$ cat src/book.txt
Book Management Module

The Book Management Module is responsible for managing books in the Online Library Management System.

Features:
1. Add new books to the library.
2. Update existing book information.
3. Search books by title, author, or book ID.
4. Delete books from the library.
5. Track book availability.

Create branch:
ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (book-feature)
$ git add src/book.txt
warning: in the working copy of 'src/book.txt', LF will be replaced by CRLF the next time Git touches it
ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (book-feature)
$ git commit -m "Add Book Management Module"
[book-feature 080e1ac] Add Book Management Module
1 file changed, 10 insertions(+)
create mode 100644 src/book.txt
ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (book-feature)
$ git branch
* book-feature
main
The file must describe the Book Management Module and contain at least three features.

Stage and commit the changes
ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (book-feature)
$ git switch main
Deletion of directory 'src' failed. Should I try again? (y/n) y
Deletion of directory 'src' failed. Should I try again? (y/n) n
M library_info.txt
Switched to branch 'main'
ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (main)
$ |
```

```
MINGW64/c:/Users/ADITYA/OneDrive/desktop/Kushagra-Gla-LibrarySystem
ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (library-update)
$ nano library_info.txt
ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (library-update)
$ git diff
warning: in the working copy of 'library_info.txt', LF will be replaced by CRLF the next time Git touches it
diff --git a/library_info.txt b/library_info.txt
index 4d8ed34..22f4ab6 100644
--- a/library_info.txt
+++ b/library_info.txt
@@ -1,3 +1,10 @@
+Name : Kushagra
+Roll Number: 2415001887
+Subject : Application Build and Release Automation lab
+Project Name : Online Lib management system
+Semester: 5th sem
+Project Name: Advanced Online Library management system
+Project Objective: To develop an efficient Online Library
+Management System for managing books, users and library operations.
+Project Description: the system will
+provide functionalities for managing library books, tracking book availability and maintaining library
+related information.
ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (library-update)
$ git add library_info.txt
warning: in the working copy of 'library_info.txt', LF will be replaced by CRLF the next time Git touches it
ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (library-update)
$ git commit -m "Update project name in library-update"
[library-update 79a3b04] Update project name in library-update
1 file changed, 8 insertions(+), 3 deletions(-)
Git should show something similar to:
ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (library-update)
$ git switch main
Switched to branch 'main'
ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (main)
$ nano library_info.txt
ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (main)
$ git add library_info.txt
ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (main)
$ git merge library-update
Auto-merging library_info.txt
CONFLICT (content): Merge conflict in library_info.txt
Automatic merge failed; fix conflicts and then commit the result.
This is your conflict demonstration.
ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (main)
$
```

```
MINGW64/c/Users/ADITYA/OneDrive/desktop/Kushagra-Gla-LibrarySystem
Roll Number: 2415001887
Subject : Application Build and Release Automation Lab
<<<<<< HEAD
Project Name :GLA Online Library Management System
Semester: 5th sem

Project Name :Advanced Online Library management system
Semester: 5th sem
Project Objective: To develop an efficient online Library
management System for managing books, users and library operations.
Project Description: the system will
provide functionalities for managing library books, tracking book availability and maintaining library-
related information.
>>>>>> library-update

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (main/MERGING)
$ nano library_info.txt
$ git add library_info.txt
$ git status
On branch main
All conflicts fixed but you are still merging.
(use 'git commit' to conclude merge)

Changes to be committed:
  modified:   library_info.txt

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (main/MERGING)
$ git commit -m "Resolve merge conflict in library_info.txt"
(main e755a8f) Resolve merge conflict in library_info.txt
$ cat library_info.txt
Name : Kushagra
Roll Number: 2415001887
Subject : Application Build and Release Automation Lab <<<<<< HEAD
Project Name :GLA Online Library Management System
Semester: 5th sem

Project Name :Advanced Online Library management system
Semester: 5th sem
Project Objective: To develop an efficient Online Library
management System for managing books, users and library operations.
Project Description: the system will
provide functionalities for managing library books, tracking book availability and maintaining library-
related information.>>>>>> library-update

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (main)
$ git status
On branch main
nothing to commit, working tree clean
```

STEP 11 — Display final file

STEP 12 — Final Git status

Screenshot 14: final resolved library_info.txt

Screenshot 15: final git status

```
Project Description: the system will
provide functionalities for managing library books, tracking book availability and maintaining library-
related information.
>>>>>> library-update

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (main/MERGING)
$ nano library_info.txt
$ git add library_info.txt
$ git status
On branch main
All conflicts fixed but you are still merging.
(use 'git commit' to conclude merge)

Changes to be committed:
  modified:   library_info.txt

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (main/MERGING)
$ git commit -m "Resolve merge conflict in library_info.txt"
(main e755a8f) Resolve merge conflict in library_info.txt
$ cat library_info.txt
Name : Kushagra
Roll Number: 2415001887
Subject : Application Build and Release Automation Lab <<<<<< HEAD
Project Name :GLA Online Library Management System
Semester: 5th sem

Project Name :Advanced Online Library management system
Semester: 5th sem
Project Objective: To develop an efficient Online Library
management System for managing books, users and library operations.
Project Description: the system will
provide functionalities for managing library books, tracking book availability and maintaining library-
related information.>>>>>> library-update

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (main)
$ git status
On branch main
nothing to commit, working tree clean

ADITYA@Kush MINGW64 ~/OneDrive/desktop/Kushagra-Gla-LibrarySystem (main)
$ git log --oneline --graph --all
```

Viva ke liye important commands

Teacher agar pooche "ye command kya karti hai?", ye yaad rakhna:

Linux

Present Working Directory show karta hai.

Current directory ke files/folders list karta hai.